

The effect of root canal treatment on oral health-related quality of life: Clinical trial

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INTRODUCTION

Natural dentition must always be preserved because nothing appears, feels, or functions like a natural tooth.^[1] There has been a growing interest in assessing the impact of health on the quality of life to help understanding the effects of health problems and how treatment may change the lives of patients.^[2,3] The oral health impact profile (OHIP), designed by Slade and Spencer in 1994, is a well-established tool for assessing oral health-related

quality of life (OHRQoL).^[4] OHRQoL considered being an integral part of the evaluation of the effects of oral problems and the assessment of clinical outcomes.^[2,3] OHIP questionnaire, which is made to measure seven concepts, which are as follows: functional limitation, physical pain, psychological discomfort, physical disability, psychological disability, social disability, and handicap. This measurement aims to give a good assessment of the oral condition and needs.^[5] Since then, OHIP-14 has been built as a short form, consisting of 14 statements that basically record

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the critical aspects of the complete version of OHIP.^[6] In the scope of endodontics, researches used the short-form OHIP-14 and have shown that pulpal disease may have an influence on the reported OHRQoL.^[3,7,8] In fact, the aspects of physical pain have been found to influence the patient perception of their OHRQoL. In 6-month follow-up, the researchers were able to show the improvement of OHRQoL after the root canal procedures.^[8] There was a systemic review that discussed pain severity and prevalence after, before, and during root canal treatment. They found that the pain associated with pulpal involvement was severe and declined to moderate within 1 day and reduced to minimal after 7 days.^[9] Considering the limited knowledge on the influence of endodontic treatment on the everyday lives of patients, the goal of this research was to measure the impact of root canal treatment on OHRQoL in one visit relative to root canal treatment at multiple visits.

MATERIAL AND METHODS

This study was established in a dental teaching hospital, Makkah, Saudi Arabia, from the beginning of September 2019 to the end of January 2020. The institutional review board (IRB) approved the study protocol (IRB number: 132-19). The present study is a clinical trial, in which participants received root canal treatment in single or multiple visits, and OHRQoL was measured by asking the subjects to complete an Arabic version of OHIP-14 questionnaire [Table 1].

To detect a difference in mean changes of 8.0 in OHIP scores with a standard deviation (SD) of 6.0 with 82% power and alpha of 0.05, sample sizes of 16 subjects per group were included.^[10] Consequently, a convenience sample of 32 patients was included; age ranged from 16 to 60 years and all needed root canal treatment. To reduce selection bias, mixed-effect models were used. Patients diagnosed with symptomatic irreversible pulpitis and

normal periapical tissue were included in the study. Cases with calcified canals, systemic diseases that may complicate root canal treatment such as endocarditis, or any kind of resorption were excluded from the study.

After consent form was signed and research purposes were clarified to the participant, an interview form was created and followed to reduce information bias. To quantify OHRQoL and to verify sociodemographic factors, the questionnaire was delivered through face-to-face interview. Each question in the OHIP-14 has a code: (zero mean never, one mean hardly ever, two mean occasionally, three mean fairly often, and four mean very often). OHIP-14 score was calculated by summing up the scores of 14 questions, ranging from 0 to 56, then using it as an indicator of the severity of the impact on OHRQoL. The higher the score, the greater the negative impact.^[11] The patients were assigned randomly into two groups of fifty patients each. In the single-visit group, patients filled out the OHIP questionnaire twice, at the beginning of the treatment, and at the second follow-up visit after obturation. For the multiple visits group, the OHIP questionnaire was filled three times, in the beginning, the next visit after access cavity and pulp extirpation, and after obturation.

Local anesthesia (articaine 4% with epinephrine 1/100,000, Septodont, saint-maur-des-fossés, France) was administrated. Then, conventional endodontic treatment was performed for indicated tooth by undergraduate dental students in a single or multiple visits. Rubber dam medium thickness (Dentsply Sirona, Ballaigues, Switzerland) was used to isolate the indicated tooth. The working length was determined with an apex locator (DentaPort ZX Apex Locator, MORITA, Kyoto City, Japan) by a size 10 K-file, and a periapical X-ray was taken to confirm the working length. Irrigation was performed by a disposable plastic syringe and a 30G endodontic needle with 5% sodium hypochlorite. Root canal preparation was performed by

Table 1: Oral health impact profile-14 questionnaire in English and Arabic

English Version	Arabic Version
1. Had trouble pronouncing any words?	١- هل كان لديك مشاكل في لفظ أي كلمة بسبب مشاكل في أسنانك؟
2. Felt sense of taste has worsened?	٢- هل شعرت مرة أن حاسة الذوق ساءت بسبب مشاكل في أسنانك؟
3. Had painful aching?	٣- هل كان لديك أي ألم في أسنانك؟
4. Found it uncomfortable to eat any foods?	٤- هل شعرت مرة بصعوبة في تناول بعض المأكولات بسبب مشاكل في أسنانك؟
5. Been self-conscious?	٥- هل كنت واعيًا بالمشاكل في أسنانك؟
6. Felt tense?	٦- هل شعرت مرة بالتوتر بسبب مشاكل في أسنانك؟
7. Felt diet has been unsatisfactory?	٧- هل حميتك الغذائية أصبحت غير مرضية بسبب مشاكل في أسنانك؟
8. Had to interrupt meals?	٨- هل اضطرت مرة للتوقف عن الطعام بسبب مشاكل في أسنانك؟
9. Found it difficult to relax?	٩- هل وجدت مرة صعوبة في الاسترخاء بسبب مشاكل في أسنانك؟
10. Been a bit embarrassed?	١٠- هل شعرت مرة بالأحراج بسبب مشاكل في أسنانك؟
11. Been a bit irritable?	١١- هل شعرت مرة بالانزعاج مع باقي الناس بسبب مشاكل في أسنانك؟
12. Had difficulty doing usual jobs?	١٢- هل وجدت مرة صعوبة في إنجاز واجباتك الاعتيادية بسبب مشاكل في أسنانك؟
13. Felt life less satisfying?	١٣- هل شعرت مرة أن الحياة بشكل عام أقل إرضاءً بسبب مشاكل في أسنانك؟
14. Been totally unable to function?	١٤- هل كنت غير قادر كليًا على العمل بسبب مشاكل في أسنانك؟

the ProTaper NEXT nickel–titanium rotary system (X Smart Plus, Dentsply Sirona, Ballaigues, Switzerland). Ethylenediaminetetracetic acid was used as a conditioner for the root canal (Dentsply sirona, Ballaigues, Switzerland). Root canal filling was completed with gutta-percha points (Dentsply sirona, Ballaigues, Switzerland) and resin-based root canal sealer (ADSEAL, META BIOMED, Chungcheongbuk-do, South Korea). In multiple visit treatment, intracanal medication (Metapaste-Calcium Hydroxide with Barium Sulfate, META BIOMED, Chungcheongbuk-do, South Korea) was used between the visits. The access cavity was sealed with temporary filling between the visits. To reduce operator bias, two undergraduate students with similar performance, based on their clinical performance in the endodontics course, were selected to execute the treatments under the supervision of a single endodontist.

CONCLUSION

OHRQoL was improved after endodontic treatment, but there was no significant difference in whether the treatment was done in a single visit or multiple visits.

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Conflicts of interest

There are no conflicts of interest.

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